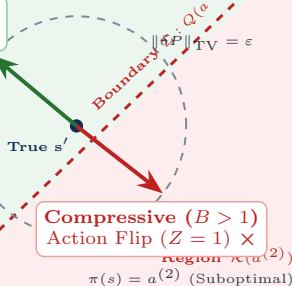


(a) Directional Geometry & Mismatch

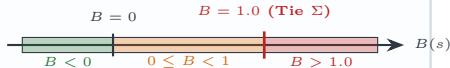
Region $\mathcal{R}(a^*)$
 $\pi^*(s) = a^*$ (Optimal)

Expansive ($B < 0$)
 Safe ($C_i = 0$) ✓



Objective Mismatch: Identical $\|\delta P\|_{TV}$ error yields opposite outcomes ($C_i = 0$ safe vs. $Z = 1$ flip).

(b) Margin Gauge & Four Regimes



1. Expansive
 ($B < 0$) [✓]
 $\Delta m > 0 \Rightarrow m_{\hat{P}} > m_P$
 Optimal a^* reinforced.

2. Sub-thresh
 ($0 \leq B < 1$) [≈]
 $-\Delta m < \Delta m \leq 0 \Rightarrow m_{\hat{P}} > 0$
 Optimal a^* preserved.

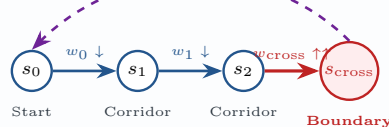
3. Tie (Σ) ($B = 1.0$) [Tie]
 $\Delta m = -m_P \Rightarrow m_{\hat{P}} = 0$
 Degenerate: $\hat{Q}(a^*) = \hat{Q}(a^{(2)})$.

4. Crossing ($B > 1.0$) [✗]
 $\Delta m < -m_P \Rightarrow \hat{Q}(a^{(2)}) > \hat{Q}(a^*)$
Action Flip:
 $Z_{\text{cross}} = 1$.

Diagnostic Metric: $B(s) = -\Delta m(s)/m_P(s)$ is the sharp causal parameter for action failure.

(c) Spatial Corridor & Propagation

Global Backup: $V(s) \leftarrow \max_a [r + \gamma EV(s')]$



1. Boundary Up-weight
 $w(s_{\text{cross}}) \uparrow (\lambda > 0)$

2. Corridor Depletion
 $w(s_1, s_2) \downarrow (\sum w_i = N)$

4. Upstream Policy Flips
 Action flips at $s_0 \downarrow$

3. Value Distortion
 $\hat{V}(s')$ backup noise

Bottleneck: Sample reallocation starves corridor dynamics, propagating errors to invert upstream actions.